

DIANA MARIN

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EDUCATION

Technische Universität Wien

September 2021 - February 2025

Doctoral Programme in Engineering Sciences

Thesis: Proximity-Based Point Cloud Reconstruction, Grade 1

Advisors: Michael Wimmer, Stefan Ohrhallinger

University of Leeds

2017 - 2021

Master of Engineering and Bachelor of Science with Honours Class I

Computer Science with High Performance Graphics and Games Engineering

PUBLICATIONS

- “*SING: Stability-Incorporated Neighborhood Graph*” - **Diana Marin**, Amal Dev Parakkat, Stefan Ohrhallinger, Michael Wimmer, Steve Oudot, and Pooran Memari. In: SIGGRAPH Asia 2024 Conference Papers (SA '24). doi: [10.1145/3680528.3687674](https://doi.org/10.1145/3680528.3687674).
- “*Reconstructing Curves from Sparse Samples on Riemannian Manifolds*” - **Diana Marin**, Filippo Maggioli, Simone Melzi, Stefan Ohrhallinger and Michael Wimmer. In: Computer Graphics Forum (Symposium on Geometry Processing 2024). doi: [10.1111/cgf.15136](https://doi.org/10.1111/cgf.15136).
- “*SIGDT: 2D Curve Reconstruction*” - **Diana Marin**, Stefan Ohrhallinger and Michael Wimmer. In: Computer Graphics Forum (Pacific Graphics 2022, Kyoto, Japan). doi: [10.1111/cgf.14654](https://doi.org/10.1111/cgf.14654).
- “*ChronoFuseGS: Multi-Temporal Gaussian Fusion with Per-Splat Persistence and Change Visualization*” - Tobias Batik, **Diana Marin**, Peter Kán, Hannes Kaufmann. To appear in: Pacific Graphics 2026 Conference Papers. arXiv: [2609.31339](https://arxiv.org/abs/2609.31339).
- “*Pole-Arina: A Privacy-Preserving Dataset and Benchmark for Static Pole Tricks*” - **Diana Marin**, Katharina Scheucher, Peter Kán. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops 2026. [CVF Open Access](#).
- “*SIGnificant Outlier Removal*” - **Diana Marin**, Filip Ilic, Stefan Ohrhallinger, Michael Wimmer. Computer Vision, Imaging and Computer Graphics Theory and Applications. VISIGRAPP 2024. Communications in Computer and Information Science, vol 2548, pp. 67–80. Springer, Cham, 2026. doi: [10.1007/978-3-032-07623-6_4](https://doi.org/10.1007/978-3-032-07623-6_4).
- “*Scan2Beams: Moving Towards Automated Modelling and Analysis of Structural Industrial Building Stock*” - Julia Reisinger, **Diana Marin**, Philipp Ruffnatscha, Peter Kán. 2025 European Conference on Computing in Construction. doi: [10.35490/EC3.2025.194](https://doi.org/10.35490/EC3.2025.194).
- “*Parameter-Free Connectivity for Point Clouds*” - **Diana Marin**, Stefan Ohrhallinger and Michael Wimmer. In Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications, 2024, Rome, Italy. doi: [10.5220/0012394900003660](https://doi.org/10.5220/0012394900003660).
- “*Visualizing Group Structure in Compound Graphs: the Current State, Lessons Learned, and Outstanding Opportunities*” - Henry Ehlers, **Diana Marin**, Hsiang-Yun Wu and Renata Raidou. In

Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications, 2024, Rome, Italy. doi: [10.5220/0012431200003660](https://doi.org/10.5220/0012431200003660).

- “*Projection-Based Structural Element Extraction from Point Clouds*” - **Diana Marin**, Julia Reisinger, Peter Kán, Hannes Kaufmann. GMP-X Proceedings, pp. 45–50, 2026. doi: [10.34726/12182](https://doi.org/10.34726/12182).
- “*Distributed Surface Reconstruction*” - **Diana Marin**, Patrick Komon, Stefan Ohrhallinger and Michael Wimmer. In: Eurographics 2024 - Posters. doi: [10.2312/egp.20241037](https://doi.org/10.2312/egp.20241037).
- “*Parameter-Free and Improved Connectivity for Point Clouds*” - **Diana Marin**, Stefan Ohrhallinger and Michael Wimmer. In: Eurographics 2023 - Posters. doi: [10.2312/egp.20231023](https://doi.org/10.2312/egp.20231023).
- “*SIG-based Curve Reconstruction*” - **Diana Marin**, Stefan Ohrhallinger and Michael Wimmer. In: Eurographics 2022 - Posters. doi: [10.2312/egp.20221013](https://doi.org/10.2312/egp.20221013).

EXPERIENCE

Technische Universität Wien

September 2024 - Present

Project Assistant (Postdoctoral Researcher since 2025) - Virtual & Augmented Reality Group

I am part of the *RE:STOCK* project, developing methods to extract finite element models of structural elements from real-life scans, that can be directly used with structural analysis tools, enabling the reuse of industrial buildings and vertical expansions. I am also part of the *PostDisaster* project, where we use Gaussian Splatting to quickly reconstruct large areas for efficient post-disaster analysis and planning.

École Polytechnique Paris

February 2024 - April 2024

Visiting Researcher

I was part of the Geomerix group, under Dr. Pooran Memari’s supervision. We developed a new neighborhood graph rooted in topological data analysis, which became a SIGGRAPH Asia paper.

Technische Universität Wien

September 2021 - September 2024

Project Assistant - Rendering and Modeling Group

I worked as part of the *Modeling the World at Scale* project, which is concerned with developing point cloud surface reconstruction algorithms that can be used in real-time applications, with a focus on real data and change detection. I contributed to the area of proximity-based connectivity and reconstruction.

University of Leeds

October 2020 - March 2021

Teaching Assistant

As a teaching assistant for the third-year Computer Graphics module, I assisted students with personal tutorials to help them better understand graphics concepts, and I marked part of the coursework.

University of Leeds

March 2020 - September 2020

Research Assistant

Under Dr. He Wang’s supervision, I worked on an interactive visualisation of laser-scanned data to make historical scenery available in museums in virtual reality, and a design tool that would help architects, using crowd simulation, assess the safety of environments in the current COVID-19 pandemic.

University of Leeds

June 2019 - September 2019

Summer Intern

During the *Topological detection of meteorological vortices* internship, I worked under Dr. Hamish Carr's supervision on finding a method to discover vortex rings in simulated meteorological data.

MD Lab Games

2015 - 2021

Co-Founder and Software Engineer

We created VR, educational and business applications with different partners (Collider Visuals), for various clients (Mercedes Benz, Procter & Gamble, Alexander McQueen and Bottega Veneta).

TEACHING AND SUPERVISION

- Programming Techniques for Visual Computing, *TU Wien*, 2024-2025.
- Computer Graphics Lab, *TU Wien*, 2022-2023.
- Teaching Assistant for Computer Graphics, *University of Leeds*, 2020-2021.
- Day-to-day supervisor (formally second supervisor) of 3 master's and 2 bachelor's theses to successful defense, with 2 more of each currently ongoing.
- Supervisor of Ole Siemers's CESC paper ([Image Based Level-of-Detail Optimization for Large-Scale 3D Reconstruction](#)), which received the **Best Paper Award** at CESC 2026.
- David Škrebčić's master's thesis *Ring-Based AR Controls for Spectacles*, which I supervised day to day (formally as second supervisor), is **nominated** for the [TU Wien Informatics Best Master Thesis Award](#) 2026.

ACHIEVEMENTS

- Selected as **WiGRAPH Rising Star** 2025 - a competitive award for outstanding early-career researchers.
- **Technical Papers Committee** member, SIGGRAPH 2027.
- **International Program Committee Member** for VMV 2025, ICVR 2026, STAG 2026.
- **Organizer and International Program Committee Member** for the Central European Seminar on Computer Graphics - CESC (2022 - present).
- **Reviewer for multiple conferences and journals**, including: Eurographics, IEEE VR, 3DV, VMV, CAGD, Graph Drawing, ICXR, Journal of Artificial Intelligence, Journal of Computational Engineering and Design.
- **Organizer of game development events**: yearly jam sites for the Global Game Jam (2018 - present), and Game Jam Plus (2023).
- **Invited talks**: FAU Erlangen, October 8, 2024: *Proximity-based reconstruction from point clouds*; TU Graz, May 28, 2024: *Sampling and reconstructing point clouds*, with Stefan Ohrhallinger.
- **KUWI Study Abroad Grant** 2024 - 1500 EUR supporting the École Polytechnique research stay.
- **Doktorandinnen der Informatik ans Rednerpult** 2022-2024 - 1000 EUR per year (3000 EUR in total) to present accepted work at a conference.
- **Cook Prize** in 2021 - awarded to the student from the School of Computing who achieves the best performance in the master's year.
- **Hutchinson Prize** in 2019 - awarded to the student from the School of Computing achieving the best performance in the second year.

- **Faculty of Engineering's Dean's List** in 2018 and 2019 - awarded to top 5% of 1st and 2nd year students in the Faculty of Engineering.
- Four-time recipient of the **Dean of Engineering Scholarship** - awarded to students achieving A*AA at A-level or equivalent and maintaining an average above 60% in each year.
- **First Place** - FinTech Hackathon, University of Leeds Hackathon and Leeds Hackathon (2018, 2019).
- **Gold Medal Infomatrix**, National and International Level, Romania, 2016.
- Exhibited at various events, showcasing the group, the research and the projects we develop: EGX and EGX Rezzed (UK, 2018) and Esports Festival Wien (2026).